

MY SQL PROJECT QUERIES USING IRIS DATASET

```
CREATE DATABASE DATA;
```

```
USE DATA;
```

```
CREATE TABLE IRIS_DATA(
```

```
    SEPAL_LENGTH FLOAT,
```

```
    SEPAL_WIDTH FLOAT,
```

```
    PETAL_LENGTH FLOAT,
```

```
    PETAL_WIDTH FLOAT,
```

```
    SPECIES VARCHAR(50)
```

```
);
```

```
SHOW DATABASES;
```

```
SHOW TABLES;
```

```
## SELECT QUERY ##
```

```
SELECT * FROM IRIS_DATA;
```

```
## FILTERING ROWS THROUGH WHERE CLAUSE ##
```

```
SELECT * FROM IRIS_DATA WHERE SPECIES = 'SETOSA';
```

```
## AVERAGE ##
```

```
SELECT AVG(SEPAL_LENGTH) AS AVG_SEPAL FROM IRIS_DATA;
```

```
## MAXIMUM
```

```
SELECT MAX(PETAL_WIDTH) AS MAX_PETAL FROM IRIS_DATA;
```

```
## MINIMUM ##
```

```
SELECT MIN(SEPAL_WIDTH) AS MIN_SEPAL FROM IRIS_DATA;
```

```
## COUNT ##
```

```
SELECT COUNT(*) AS TOTAL_ROWS FROM IRIS_DATA;
```

```
## GROUPBY ##
```

```
SELECT SPECIES, AVG(SEPAL_LENGTH) AS AVG_SEPAL_LENGTH FROM  
IRIS_DATA GROUP BY SPECIES;
```

```
SELECT SPECIES, AVG(PETAL_LENGTH) AS AVG_PETAL_LENGTH FROM  
IRIS_DATA GROUP BY SPECIES;
```

```
## SUM ##
```

```
SELECT SUM(SEPAL_LENGTH) AS TOTAL_SEPAL_LENGTH FROM IRIS_DATA;
```

```
SELECT SUM(PETAL_WIDTH) AS SUM_PETAL_WIDTH FROM IRIS_DATA WHERE  
SEPAL_LENGTH > 6;
```

```
## JOIN TABLE ##
```

```
CREATE TABLE SPECIES_INFO(  
SPECIES VARCHAR(50),  
FLOWER_COLOR VARCHAR(20));
```

```
INSERT INTO SPECIES_INFO(SPECIES, FLOWER_COLOR) VALUES  
( 'SETOSA','BLUE'),  
( 'VERSICOLOR','PURPLE'),  
( 'VIRGINICA','WHITE');
```

```
SELECT * FROM SPECIES_INFO;
```

```
SELECT I.SEPAL_LENGTH,I.PETAL_LENGTH,I.SPECIES,S.FLOWER_COLOR FROM  
IRIS_DATA I JOIN
```

```
SPECIES_INFO S ON I.SPECIES= S.SPECIES;
```

```
## ORDERBY - SORTING ##
```

```
SELECT * FROM IRIS_DATA ORDER BY PETAL_LENGTH DESC LIMIT 10;
```

```
## SUBQUERY - FIND SPECIES WITH HIGHEST AVERAGE SEPAL LENGTH ##
```

```
SELECT SPECIES FROM IRIS_DATA GROUP BY SPECIES ORDER BY  
AVG(SEPAL_LENGTH) DESC LIMIT 1;
```

```
## CREATE INDEXES (IF FILTERING/SEARCHING A COLUMN OFTEN) ##
```

```
CREATE INDEX IDX_SPECIES ON IRIS_DATA(SPECIES);
```

```
## USE LIMIT TO REDUCE THE OUTPUT SIZE ##
```

```
SELECT * FROM IRIS_DATA LIMIT 10;
```

```
## USE WHERE CLAUSE TO FILTER BEFORE GROUP BY OR JOIN ##
```

```
SELECT SPECIES, AVG(SEPAL_LENGTH) FROM IRIS_DATA WHERE  
SEPAL_LENGTH > 5 GROUP BY SPECIES;
```

MY SQL PROJECT QUERIES USING HOUSE PREDICTION

DATASET

```
CREATE DATABASE HOUSE;
```

```
USE HOUSE;
```

```
CREATE TABLE house_prediction_data (
```

```
    CRIM FLOAT,
```

```
    ZN FLOAT,
```

```
    INDUS FLOAT,
```

```
    CHAS INT,
```

```
    NOX FLOAT,
```

```
    RM FLOAT,
```

```
    AGE FLOAT,
```

```
    DIS FLOAT,
```

```
    RAD INT,
```

```
    TAX FLOAT,
```

```
    PTRATIO FLOAT,
```

```
    B FLOAT,
```

```
    LSTAT FLOAT,
```

```
    MEDV FLOAT
```

```
);
```

```
select * from house_prediction_data;
```

```
## TOTAL OF HOUSE PRICES
```

```
SELECT SUM(MEDV) AS total_house_value FROM house_prediction_data;
```

```
## TOTAL TAX COLLECTED
```

```
SELECT SUM(TAX) AS total_tax FROM house_prediction_data;
```

```
## TOTAL OF ALL NUMERIC COLUMNS ##
```

```
SELECT SUM(CRIM) AS total_crime_rate, SUM(RM) AS total_rooms, SUM(TAX) AS  
total_tax,
```

```
SUM(MEDV) AS total_value FROM house_prediction_data;
```

```
## Average House Price ##
```

```
SELECT AVG(MEDV) AS avg_price FROM house_prediction_data;
```

```
## AVERAGE PRICE BY ACCESS TO CHARLES RIVER ##
```

```
SELECT CHAS, AVG(MEDV) AS avg_price FROM house_prediction_data GROUP BY  
CHAS;
```

```
## TOP 10 HIGH-VALUE HOUSES ##
```

```
SELECT * FROM house_prediction_data ORDER BY MEDV DESC LIMIT 10;
```

```
## TOTAL COUNT BY NUMBER OF ROOMS ##
```

```
SELECT RM, COUNT(*) AS house_count FROM house_prediction_data GROUP BY RM  
ORDER BY RM;
```

```
## JOIN EXAMPLE ##
```

```
CREATE TABLE crime_info (  
    CRIM FLOAT,  
    risk_level VARCHAR(20)  
);
```

```
INSERT INTO crime_info (CRIM, risk_level)
```

```
VALUES (0.00632, 'Low'), (0.02731, 'Medium'), (0.03237, 'High');
```

```
SELECT h.CRIM, h.MEDV, c.risk_level FROM house_prediction_data h JOIN  
crime_info c ON h.CRIM = c.CRIM;
```

```
## QUERY OPTIMIZATION ##
```

```
CREATE INDEX idx_medy ON house_prediction_data(MEDV);
```

```
CREATE INDEX idx_chas ON house_prediction_data(CHAS);
```

```
SELECT AVG(MEDV) FROM house_prediction_data WHERE RM > 6;
```